

# Spoken Tutorial – Code and Audio Generation with AI

Title: Code and Audio Generation with AI

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| <b>Module</b>                 | Generative AI Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Episode</b>                | 4.3 Code and Audio Generation                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Learning Objective</b>     | At the end of this tutorial learner will be able to<br>1. Explain how <b>AI</b> generates code and audio, and describe their practical applications.                                                                                                                                                                                                                                                                                                              |
| <b>Approx. Duration</b>       | 3-4 min                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Outline</b>                | <ul style="list-style-type: none"><li>• Introduction to <b>Generative AI</b> in new domains.</li><li>• Understanding <b>Code Generation</b> as a developer's assistant.</li><li>• Exploring how <b>AI</b> suggests code and finds bugs.</li><li>• Discovering <b>Audio Generation</b> for voices and music.</li><li>• Applications of <b>Voice Synthesis</b> and <b>Text-to-Speech</b>.</li><li>• Practical uses of <b>AI</b> in <b>Music Creation</b>.</li></ul> |
| <b>Meta Tags</b>              | <b>AI, Generative AI, Code Generation, Audio Generation, GitHub Copilot, Voice Synthesis, Text-to-Speech, Music Creation, Developer Tools, Sound AI, Creative AI</b>                                                                                                                                                                                                                                                                                              |
| <b>Pre-requisite Tutorial</b> | Basic understanding of <b>Generative AI</b> concepts.                                                                                                                                                                                                                                                                                                                                                                                                             |

## Script

| Visual Cue                                                                                                        | Narration                                                                                                                                                                                              |
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| Abstract illustration representing <b>AI</b> generating code and audio waves.                                     | Welcome to this Spoken Tutorial on Code and Audio Generation with <b>AI</b> .                                                                                                                          |
| None                                                                                                              | In this tutorial, you will learn to:<br>Explain how <b>AI</b> generates code and audio.<br>Describe their practical applications.                                                                      |
| Screenshot of a web browser interface.                                                                            | To follow this tutorial, you will need a web browser.<br>We will use it to explore online <b>AI</b> tools.                                                                                             |
| None                                                                                                              | To follow this tutorial, learners should have a basic understanding of:<br><b>Generative AI</b> concepts.                                                                                              |
| Illustration showing diverse <b>AI</b> capabilities beyond text and images, like code snippets and musical notes. | So far, we have explored <b>Generative AI</b> for text and images.<br>But <b>AI</b> is mastering other complex domains.<br>Today, we will look at <b>Code Generation</b> and <b>Audio Generation</b> . |

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| <p>Illustration of a programmer typing, with <b>AI</b> suggestions appearing on the screen, possibly showing <b>GitHub Copilot</b> interface.</p> | <p>Let's start with <b>Code Generation</b>.<br/> Think of <b>AI</b> code generation as having a super-smart coding assistant.<br/> It's like having a co-pilot who knows billions of lines of code. Imagine you are typing code for a function. As you type, the <b>AI</b> suggests the next few lines. It can even complete complex logic for you.<br/> Or, you write a comment like '// function to sort a list'. The <b>AI</b> then writes the actual code instantly. This greatly speeds up the development process. It helps developers write code faster and more efficiently.</p> |
| <p>Diagram showing various ways <b>AI</b> assists coding: code completion, comment-to-code, bug detection.</p>                                    | <p>These <b>AI</b> models are trained on vast amounts of public code. They don't replace human programmers. Instead, they act as incredibly intelligent assistants. They can suggest the next few lines of code as you type. They can translate comments into functional code. And they can even help find bugs. This makes developers significantly faster and more efficient.</p>                                                                                                                                                                                                      |
| <p>Illustration of sound waves transforming from text, and musical notes being generated by <b>AI</b>.</p>                                        | <p>Now, let's move to <b>Audio Generation</b>.<br/> It's like having a digital sound wizard at your fingertips. These <b>AI</b> models are trained on sounds, voices, and music. Imagine you need a voiceover for a video. You type the script, and the <b>AI</b> generates a natural-sounding voice.<br/> Or, you need background music. You describe the mood, and the <b>AI</b> composes a unique track. This unlocks new creative possibilities for sound and music.</p>                                                                                                             |
| <p>Illustration of a speaker icon with text transforming into sound waves, representing <b>Text-to-Speech</b>.</p>                                | <p><b>Voice Synthesis</b> and <b>Text-to-Speech</b> are key capabilities. They create incredibly realistic human-like voices. This is useful for audiobooks, podcasts, or virtual assistants. Think about it. No need for human voice actors for every project.</p>                                                                                                                                                                                                                                                                                                                      |
| <p>Illustration of a music staff with <b>AI</b> algorithms generating musical notes and compositions.</p>                                         | <p><b>AI</b> can also excel in <b>Music Creation</b>. It can compose royalty-free background music for a video. It can generate entirely new melodies. Some <b>AI</b> can even create songs in the style of a famous artist. Notice how <b>AI</b> becomes a creative partner.</p>                                                                                                                                                                                                                                                                                                        |

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| None                                                                                                | <p>With this, we come to the end of this tutorial.</p> <p>Let us summarise.</p> <p>We learned that <b>Generative AI</b> extends to <b>Code Generation</b>.</p> <p>It acts as a smart assistant for developers.</p> <p>We also explored <b>Audio Generation</b>.</p> <p>This includes <b>Voice Synthesis</b>, <b>Text-to-Speech</b>, and <b>Music Creation</b>.</p>                 |
| Illustration of a person interacting with an <b>AI</b> interface for code and audio generation.     | <p>As an assignment, please do the following.</p> <p>Find an online <b>AI</b> code assistant tool.</p> <p>Experiment with generating code from comments or suggestions.</p> <p>Or, find an online <b>AI</b> audio tool.</p> <p>Try generating a short piece of music or a voice clip from text.</p> <p>Observe the quality and versatility of the <b>AI</b> generated outputs.</p> |
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